GeoCov: Geometry-Aware Fixed-Budget Memory for Long-Horizon Video Generation

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Abstract

When a video generator revisits a scene, its memory must supply useful earlier views. Retaining all history preserves those views but leaves their selection to an imperfect retriever; recency and appearance-based curation can instead discard useful evidence or retain evidence the retriever overlooks. We study these retention and selection errors and propose a design principle: preserve view coverage by removing observations with geometric and visual substitutes. Geometric Coverage (GeoCov) implements this principle through an online pose–appearance graph with a fixed memory budget. It requires no training or changes to the generator and retriever. On fifteen matched 180-second MemCam trajectories, GeoCov-32 stores 32 rather than 5,397 frames while reducing LPIPS from 0.5980 to 0.5876 and FVD from 734.2 to 476.6. On fifteen 60-second WorldMem videos, a corresponding geometric policy reduces LPIPS from 0.652 to 0.534 and FVD from 3077.6 to 1116.9. Controls with shared candidate pixels show that curated archives yield cleaner selected memories, while random history expansion provides no evidence of a candidate-count penalty. These results support constructing bounded memory around view coverage and retrieval.

1 Introduction

A camera leaves a room and returns minutes later. To reconstruct the same objects, a video generator needs memories of that room, even if its recent context depicts somewhere else. The memory must preserve relevant earlier views and supply them when needed. These are distinct requirements: a useful observation can be deleted before a revisit, or remain stored but be overlooked by the retriever.

Existing approaches illustrate two broad memory designs. Context-as-Memory and MemCam retrieve frames by camera field-of-view (FOV) overlap from accumulated history (Yu et al., 2025; Gao et al., 2026). Other approaches construct a compact context through explicit selection or compression rules. FramePack allocates context according to frame importance, including temporal proximity and feature similarity (Zhang et al., 2025); MemFlow combines prompt-relevant memories with a representative frame from the preceding chunk (Ji et al., 2025). These designs operate at different stages: a bounded conditioning context can still be read from a growing persistent archive.

Retaining all history solves availability but leaves selection unresolved. Storage grows with the rollout, and exhaustive retrieval searches an increasing number of candidates. More subtly, the archive contains self-generated observations. A plausible camera pose can point to an image whose scene content has drifted. In our complete-retention rollouts, later retrieved indices are better aligned with the target view, yet the generated images at those indices are more corrupted (Fig. 4). Keeping a useful observation somewhere in history therefore does not ensure that it conditions the next chunk.

Selective construction faces the complementary problem: the criterion used to retain memories may not preserve the evidence needed at a future camera revisit. Recency favors local continuation, while appearance diversity favors distinctive content. We isolate these criteria with FIFO and an appearance-based rarity–irreplaceability baseline (RI) inside the same generator. FIFO incurs a large loss of useful history; RI preserves much more of it, but leaves a larger gap between its retrieved and best retained item than GeoCov (Fig. 1). These controlled comparisons identify a failure mode of the tested criteria: preserving recent or diverse observations alone does not ensure that useful views remain retrievable.

These observations motivate a memory design organized around *view coverage and substitutability*. A frame should remain valuable when few other observations cover its view, even if it is old or visually ordinary. It becomes a candidate for eviction when several nearby observations provide similar visual evidence. This adapts the coverage intuition behind SLAM keyframe culling (Mur-Artal et al., 2015; Mur-Artal and Tardos, 2017) to self-generated memories that condition a diffusion model.

We implement this principle as *Geometric Coverage* (GeoCov). At each update, GeoCov combines existing and new memories in a pose–appearance graph, scores each observation by neighbor support and its closest substitute, and evicts low-utility items until B remain. Persistent archive size and exact candidate scoring per query are independent of rollout length at fixed B. The video model, denoising schedule, and retrieval rule remain unchanged.

On fifteen matched 180-second MemCam trajectories, GeoCov-32 retains 32 of 5,397 frames while reducing

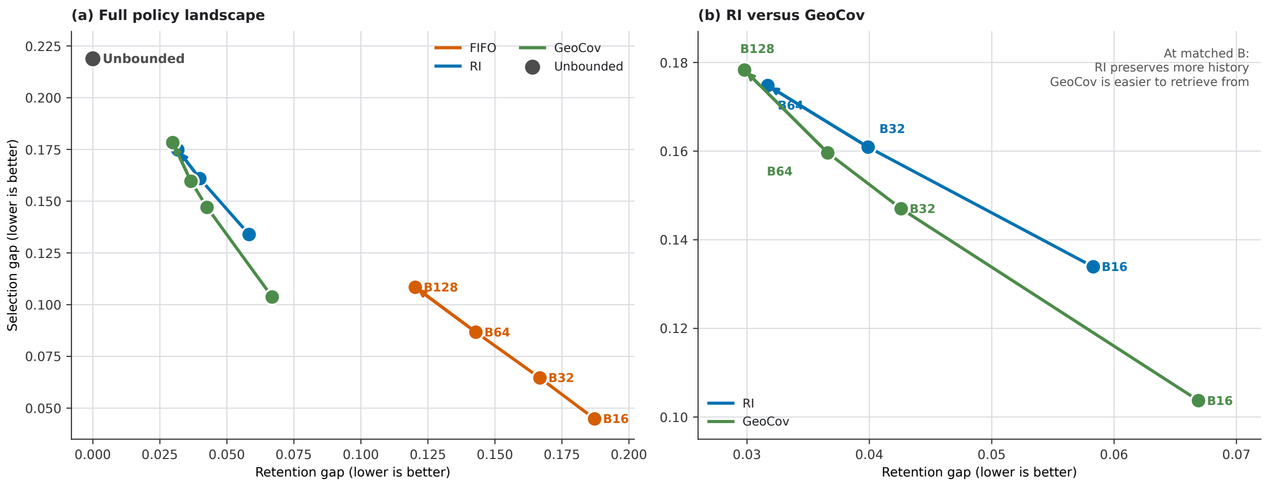

Figure 1: Memory must preserve evidence and make it retrievable. Retention gap measures useful history lost to deletion; selection gap measures the retriever's shortfall from the best retained item. Complete retention has no deletion loss but the largest selection gap; FIFO loses useful history. At matched budgets, RI has lower retention gap than GeoCov, while GeoCov has lower selection gap. Values use common-source pixels on thirteen trajectories; lower-left is better.

LPIPS from 0.5980 to 0.5876 and FVD from 734.2 to 476.6. It also outperforms the tested FIFO and RI archives on both metrics. A corresponding controller improves LPIPS from 0.652 to 0.534 and FVD from 3077.6 to 1116.9 on fifteen 60-second WorldMem videos. These results test the design in two generation systems. Controls explain its behavior: with pixels shared across policies, curated candidate sets yield cleaner selected indices; randomly expanding a fixed recent-history pool does not systematically lower their fidelity. Together, the evidence supports archive composition as a design variable in long-horizon generation.

Our contributions are:

- a retention–selection analysis that identifies complementary failures of complete retention and the tested memory-construction rules;
- an online controller that uses geometric and visual substitutes to maintain view coverage within a fixed archive budget;
- matched quality results in MemCam and WorldMem, with shared-pixel and fixed-history controls that test the role of archive composition.

2 Problem Setting

At chunk t, let P_t denote the known camera trajectory, M_{t-1} the persistent archive, and N_t the memories produced in the new chunk. A memory-guided generator follows

$$C_t = R(P_t; M_{t-1}), \tag{1}$$

$$\hat{Y}_t = G(\hat{Y}_{t-1}, P_t, C_t), \tag{2}$$

$$M_t = U(M_{t-1}, N_t; B), \tag{3}$$

where R retrieves a small context, G generates the next chunk, and U updates the archive. Complete retention appends N_t and ignores B. Our intervention changes only U.

Suppose each of T chunks contributes s candidates. Complete retention stores $O(Ts)$ items. Exhaustive retrieval scores $O(ts)$ items at chunk t and performs $O(T^2 s)$ comparisons over the rollout. A fixed archive stores $O(B)$ items and requires $O(B)$ exact comparisons per query, or $O(TB)$ over the rollout. These are properties of the evaluated exact implementation, not lower bounds on retrieval: approximate nearest-neighbor or pose indices can accelerate either archive (Johnson et al., 2017).

Complete retention is an availability oracle because it never deletes an item. It is not a quality oracle: the retriever can overlook a useful item or select a self-generated observation that has drifted. A bounded controller must therefore balance two errors. Deleting too aggressively removes useful history, while an uncurated archive can present the fixed retriever with a poorly organized candidate set.

3 Geometric Coverage Memory

GeoCov operationalizes the two requirements in Fig. 1: retain representatives of sparsely covered views, and remove redundant alternatives where several substitutes are available. We estimate substitutability using camera pose and visual appearance. This gives a pose–appearance proxy for shared view coverage without requiring a reconstructed scene.

At each update, GeoCov forms the prospective archive $V_t = M_{t-1} \cup N_t$. Item $i \in V_t$ has camera-to-world pose (R_i, t_i) and a normalized DINOv2-Base descriptor z_i (Oquab et al., 2024). Let $\widetilde{d}_t$ be the median nonzero pairwise translation distance in V_t, with value one if all positions coincide. We define

$$d_p(i,j) = \frac{\|t_i - t_j\|_2}{\max(\widetilde{d}_t, 10^{-6})} + 2\frac{\theta(R_i^\top R_j)}{\pi}, \tag{4}$$

$$\theta(R) = \arccos\left(\operatorname{clip}\left(\frac{\operatorname{tr}(R)-1}{2}, -1, 1\right)\right), \tag{5}$$

$$K(i,j) = 0.65\exp[-d_p(i,j)] + 0.35\max(z_i^\top z_j, 0). \tag{6}$$

Pose receives most of the affinity weight; DINO separates nearby observations whose visible content differs.

Let

$$c_i = \sum_{j\neq i} \mathbb{I}[K(i,j) \geq 0.65], \qquad k_i^{\max} = \max_{j\neq i} K(i,j). \tag{7}$$

The retention utility is

$$u_i^{\text{Geo}} = 1 - \min(c_i/3, 1) + \frac{0.5}{c_i+1} + 0.25(1-k_i^{\max}). \tag{8}$$

The first two terms favor sparsely covered regions. The final term favors an item without a close substitute. Low utility therefore identifies local redundancy in camera and appearance space. The controller removes the lowest-utility eligible items until $|M_t| \leq B$. Its direct implementation builds an $O((B+s)^2)$ affinity matrix and stores $O(B)$ persistent items.

Protected items. The reported MemCam implementation reserves the initial conditioning frame in GeoCov and RI, while complete retention contains it by construction. FIFO and K-center do not reserve it. All bounded policies temporarily protect the newest section endpoint during an update so that the next chunk has current local context. Protected items count toward B. Section 5.5 quantifies the initial-frame contribution; no future view or ground truth is used by the controller.

4 Experimental Setup

MemCam. We use the public MemCam implementation based on the Wan2.1 1.3B diffusion transformer (Team, 2025; Gao et al., 2026). The primary evaluation contains fifteen matched 180-second trajectories from the Context-as-Memory dataset (Yu et al., 2025). Each rollout contains 5,397 generated frames in 76-frame target sections with 50 denoising steps. Policies share checkpoints, prompts, camera paths, and seeds. The primary budget is $B = 32$; we also test $B \in \{16, 64, 128\}$.

WorldMem. We evaluate the first fifteen matched 60-second WorldMem videos using its latent-memory interface (Xiao et al., 2025). The geometric controller is adapted to the stored representation while keeping the same view-coverage principle. Results are compared only within WorldMem.

Baselines. Complete retention keeps the full history. FIFO retains the most recent B items. K-center maintains descriptor coverage. Marginal Coverage Eviction (MCE) approximates a facility-style objective. Rarity–irreplaceability (RI) clusters DINO features and favors small appearance clusters whose members lack an RGB substitute. These policies test full availability, recency, generic diversity, marginal coverage, and appearance rarity. All use the original system retriever after archive construction.

Metrics. LPIPS (Zhang et al., 2018) compares generated frames with trajectory-indexed ground truth. FVD (Unterthiner et al., 2018) uses the same StyleGAN-V I3D detector, 16-frame clips, four clips per video, stride four, and 224-pixel inputs within each comparison. Standard VBench (Huang et al., 2024) measures subject and background consistency, motion smoothness, dynamic degree, aesthetics, and imaging quality. PSNR, SSIM (Wang et al., 2004), and DINO distance (Oquab et al., 2024) are used only for memory diagnostics. Diagnostic confidence intervals bootstrap trajectory-level means.

Evaluation scope. The policy family and constants were developed while inspecting the MemCam suite, so those trajectories are not an untouched test set. WorldMem changes the backbone and memory interface but was evaluated after MemCam. Headline LPIPS, FVD, and VBench values are matched point estimates over fifteen videos; we do not claim statistical significance for them. CUT3R (Wang et al., 2025) is excluded after its evaluator failed a ground-truth camera sanity check; VBench-Long is excluded because its evaluation is incomplete.

5 Results

5.1 Long-horizon video quality

Table 1: Matched 180-second MemCam results. All bounded policies use $B = 32$; lower is better.

Policy	Stored	LPIPS ↓	FVD ↓
Complete retention	5,397	0.5980	734.2
FIFO	32	0.6514	677.3
RI	32	0.5939	550.4
GeoCov	32	**0.5876**	**476.6**

Table 1 presents the primary result. Relative to complete retention, GeoCov reduces FVD by 35.1% and LPIPS by 0.0104 while retaining 0.59% of the frames. RI also improves both metrics but remains behind GeoCov. FIFO improves FVD and substantially worsens LPIPS, showing that the tested gain does not follow from using a small recent-history bank.

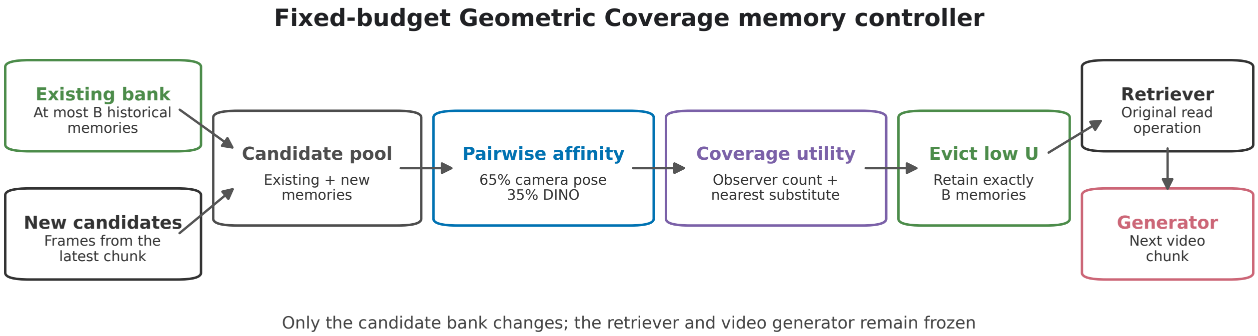

Figure 2: Online GeoCov update. Existing and new memories form a pose–appearance graph. GeoCov evicts locally redundant nodes to meet the budget; the original retriever and generator then use the curated bank.

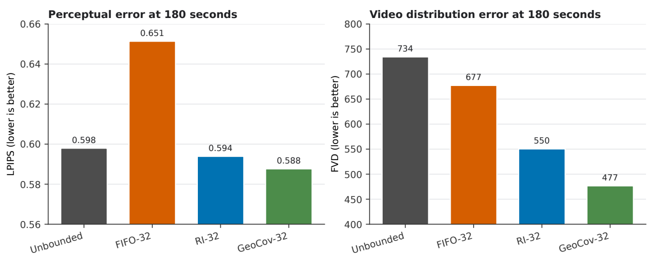

Figure 3: MemCam quality at 180 seconds. Structured B32 archives improve LPIPS and FVD over complete retention; FIFO does not reproduce the same result.

Table 2: Matched 60-second WorldMem results at $B = 32$. Values are comparable only within WorldMem; lower is better.

Policy	LPIPS ↓	FVD ↓
Complete retention	0.652	3077.6
FIFO	0.689	3554.9
Latent-RI	0.546	1160.4
Geometric Coverage	**0.534**	**1116.9**

The same ordering appears in WorldMem (Table 2). Geometric Coverage improves LPIPS by 0.118 and FVD by 1960.7 relative to complete retention. This result supports the bounded geometric principle under a second backbone and memory interface; it does not imply that the exact MemCam score is universally optimal.

5.2 Budget sensitivity

Table 3: GeoCov across tested budgets. MemCam values use 180-second rollouts; WorldMem values use 60-second videos. Lower is better.

	MemCam		WorldMem
B	LPIPS	FVD	LPIPS
16	0.5876	**446.1**	**0.525**
32	0.5876	476.6	0.534
64	**0.5865**	493.9	0.545
128	0.5913	515.5	0.577

The result is not isolated to B32 (Table 3). MemCam LPIPS is stable from B16 through B64, and every tested GeoCov budget improves FVD over complete retention. WorldMem LPIPS also remains better than complete retention through B128. Larger is not consistently better within this policy family, so we treat B as a system parameter rather than claim a universal optimum.

5.3 Testing coverage and retrieval

For target q, let $d(i, q)$ be the DINO distance between generated memory i and exact-index target ground truth. Let $\mathcal{H}$ be complete history, M a policy archive, $i^*_M = \arg\min_{i \in M} d(i, q)$ the hindsight-best retained item, and i_R the item chosen by the real retriever. We define

$$\Delta_{\text{ret}} = d(i^*_M, q) - \min_{i \in \mathcal{H}} d(i, q), \tag{9}$$

$$\Delta_{\text{sel}} = d(i_R, q) - d(i^*_M, q). \tag{10}$$

Retention gap measures quality lost by deletion. Selection gap measures quality present in the archive but unused by the retriever. Both require ground truth and are diagnostics, not runtime objectives.

Table 4: Common-source gaps at B32 on thirteen trajectories. Lower is better.

Policy	Retention gap	Selection gap
Complete retention	0.0000	0.2188
FIFO	0.1668	**0.0646**
RI	**0.0399**	0.1609
GeoCov	0.0426	0.1470

Complete retention has zero retention gap by definition but the largest selection gap (Table 4). FIFO creates an easy selection problem by discarding distant history, at the cost of the largest retention gap. RI and GeoCov preserve substantially more useful history while reducing the selection gap relative to complete retention. At B32, GeoCov gives up 0.0027 retention gap relative to RI and gains 0.0139 selection gap.

Every measured increase from B16 to B128 lowers retention gap and raises selection gap (Fig. 1). For Geo-

Cov, their sum is 0.1706, 0.1896, 0.1962, and 0.2081 at B16, B32, B64, and B128, respectively, versus 0.2188 for complete retention. The archive can contain better evidence without making that evidence easier for a fixed retriever to use.

5.4 Common-source selection control

Independent policy rollouts contain different pixels because an early memory choice changes later generation. To isolate archive composition, we reconstruct each policy's candidate set and selected indices but load every candidate image from the same complete-retention rollout. Each selected image is compared with ground truth at its own historical index. Thus the pixels are fixed; only the candidate sets and selected identities differ.

Table 5: Common-source selected-frame fidelity on fifteen trajectories. Deltas are relative to complete retention; positive is better.

Selector	PSNR	ΔPSNR	ΔSSIM
Complete retention	11.703	–	–
FIFO	11.487	-0.216	-0.0109
RI	13.478	+1.775	+0.0672
GeoCov	**16.332**	**+4.629**	**+0.1512**

GeoCov improves PSNR by 4.629 dB (95% CI $[3.521, 5.874]$) and SSIM by 0.1512 (CI $[0.1100, 0.1929]$), with positive differences on all fifteen trajectories. RI improves PSNR by 1.775 dB (CI $[1.005, 2.683]$) and SSIM by 0.0672 (CI $[0.0456, 0.0926]$). FIFO is worse. Because the source pixels are shared, these differences arise from which historical indices the unchanged retriever selects from each candidate set.

The temporal analysis in Fig. 4 helps explain what the retriever encounters. Across 4,200 queries, selected historical indices become more view-aligned from early to late rollout: DINO view mismatch changes by -0.0335 (CI $[-0.0511, -0.0171]$). Yet corruption of the generated content at those indices increases by 0.0873 (CI $[0.0236, 0.1572]$), and effective mismatch to the target increases by 0.0732 (CI $[0.0165, 0.1296]$). The hindsight-best archive mismatch remains approximately flat. This analysis is descriptive because archive composition, observation age, and rollout time co-vary.

5.5 Controls on initial-frame use and archive size

The initial conditioning frame is the only ground-truth image stored in the MemCam archive. Both GeoCov and complete retention contain it, but curation can change whether the retriever ranks it first. Among 9,927 queries where their selections disagree, GeoCov chooses the initial frame 1,483 times (14.9%). Removing those cases leaves a gain of 1.060 dB PSNR (CI $[0.213, 2.290]$) and 0.0449 SSIM (CI $[0.0079, 0.0900]$). On the stricter subset where neither selection is the initial frame and both have logged FOV overlap of at least 0.80, the difference is 0.103 dB (CI $[-0.202, 0.407]$) and 0.0098 SSIM (CI $[-0.0029, 0.0258]$). Thus initial-frame retrieval explains part, but not all, of the aggregate common-source result; the high-overlap non-initial subset is not statistically distinguishable from zero.

Table 6: Fixed-history candidate-count intervention. Winner change is relative to the shared B32 core.

Pool	PSNR ↑	SSIM ↑	Winner change
B32	11.275	0.3064	0.000
B64	11.348	0.3025	0.397
B128	11.322	0.3013	0.497
B256	11.310	0.3012	0.581
B512	11.330	0.3023	0.611
B1024	11.436	0.3068	0.661
All (3,797 mean)	11.502	0.3075	0.722

The common-source policies differ in both composition and size, so we isolate cardinality with fixed generated history and nested candidate pools. Each query starts from the same recent B32 core; older frames are added in four random orders while candidate pixels, camera poses, targets, and FOV scores remain fixed. Full history changes the B32 winner in 72.2% of queries but changes PSNR by +0.228 dB (CI $[-0.109, 0.664]$) and SSIM by +0.0011 (CI $[-0.0117, 0.0188]$). Candidate count strongly affects identity, but this intervention provides no evidence that count alone lowers fidelity. Combined with Table 5, it points to structured composition.

6 Related Work

Long-horizon video memory. Context-as-Memory stores historical RGB frames and retrieves them by camera FOV overlap (Yu et al., 2025). MemCam adds context compression and provides an open implementation of RGB-frame retrieval (Gao et al., 2026). WorldMem stores visual memories with pose and time state and reads them through learned memory attention (Xiao et al., 2025). VMem indexes prior views with surfels (Li et al., 2025); Spatia maintains an updatable point cloud (Zhao et al., 2025); Mirage stores spatial evidence in diffusion latent space (Wang et al., 2026a); and AnchorWeave retrieves local geometric memories (Wang et al., 2026b). We study the persistent candidate archive that precedes these bounded reads.

Selective context and keyframes. FramePack allocates a fixed context length using frame-importance-based compression (Zhang et al., 2025). MemFlow combines semantic retrieval with a prototype from the preceding chunk and sparse token activation (Ji et al., 2025). We target persistent archive composition under a camera-driven retriever; our FIFO and RI comparisons isolate re-

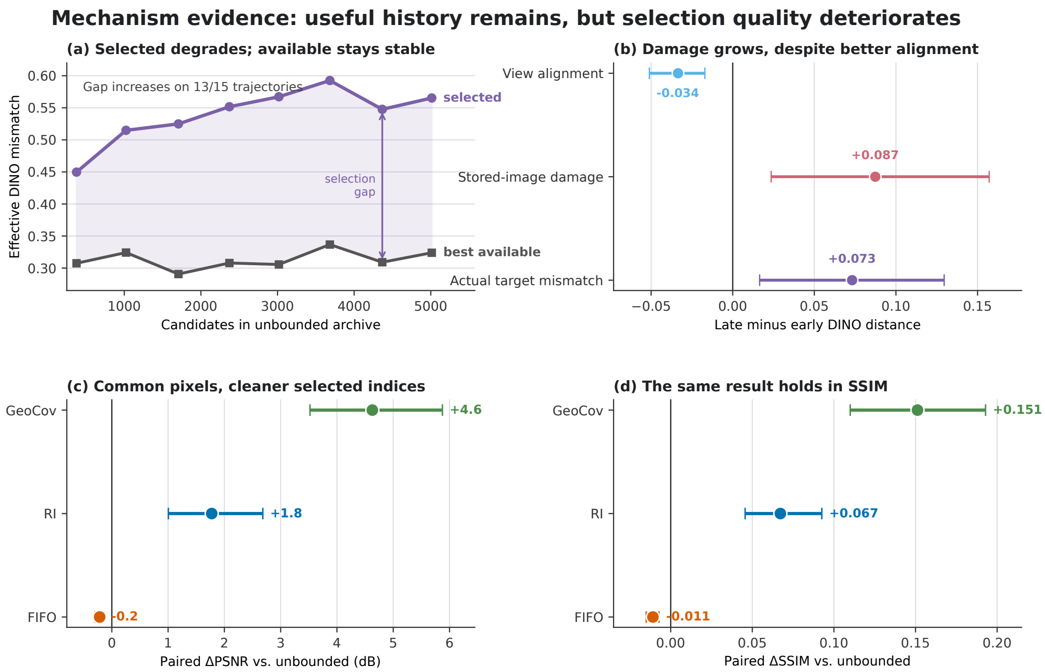

Figure 4: Selected-memory evidence. Complete-retention selections move away from the hindsight-best available item as their generated content degrades (a–b). With candidate pixels fixed, structured archives lead the retriever to cleaner historical indices (c–d). Error bars are trajectory-bootstrap 95% intervals.

cency and appearance criteria within that setting rather than reproduce these full systems. SLAM systems use co-visibility graphs and keyframe culling to remove redundant observations while preserving a map (Mur-Artal et al., 2015; Mur-Artal and Tardos, 2017). GeoCov adapts this principle to recurrent video generation, where observations are self-generated and memory conditions a diffusion model rather than a geometric optimizer. Surprise Forcing uses novelty, feedback-controlled admission, replacement, and routing (Shi et al., 2026). Streaming subset selection supplies broader coverage objectives (Nemhauser et al., 1978; Badanidiyuru et al., 2014). Our baselines include recency, K-center, appearance rarity, and marginal coverage.

7 Discussion and Limitations

The design implication is that archive construction should account for both what evidence survives and what the retriever selects. Complete retention maximizes availability; the tested recency and rarity criteria make different tradeoffs between retention and selection. GeoCov uses view coverage and substitutability to balance these requirements, improving the reported quality metrics in two systems while bounding the archive. The shared-pixel and fixed-history controls support the role of composition in those selections.

The evidence does not show that GeoCov detects corrupted images. Its score uses pose and appearance redundancy, not a calibrated quality estimate. The common-source analysis also does not prove that selected-frame fidelity causes the full FVD difference; a multi-trajectory content-replacement replay would test that link more directly. Because initial-frame reservation differs across some bounded baselines, symmetric pin/no-pin closed-loop runs remain an important control.

Finally, the MemCam constants were developed on the reported suite, the sample contains fifteen trajectories, and headline metrics lack matched uncertainty. WorldMem provides a second backbone but not an untouched development process. We have not measured matched end-to-end latency or peak memory, and the long-duration resource values in Appendix D are extrapolations. A neutral reservoir baseline, frozen evaluation on new trajectories, and FVD estimates over a larger sample would strengthen the empirical claim.

8 Conclusion

An effective video memory must preserve useful views and make them available to the retriever. Our analysis identifies retention and selection failures in complete-history, recency, and appearance-based archives. GeoCov addresses these requirements through geometric and visual substitutability: sparsely covered views remain valuable, while redundant observations are removed to meet a fixed budget. It improves the reported MemCam and WorldMem quality metrics through the original generation and retrieval interfaces. The results support view coverage as a practical principle for constructing long-horizon video memory.

A Video-Quality Ablations

Table 7: Matched 60-second MemCam ablation at $B = 32$. Lower is better for LPIPS and FVD; higher is better for the VBench dimensions.

Policy	LPIPS-10	FVD-10	LPIPS-60	FVD-60	Subject	Background	Motion	Dynamic	Aesthetic	Imaging
GeoCov	0.4964	719.3	0.5850	690.5	**0.8187**	**0.9071**	0.9914	1.000	**0.4792**	**0.5294**
Rarity only	**0.4878**	**682.2**	0.5986	732.3	0.8008	0.8996	**0.9925**	1.000	0.4495	0.4976
0.75 Geo + 0.25 rarity	0.4927	713.1	0.5872	**676.4**	0.8104	0.9006	0.9922	1.000	0.4554	0.5027
0.75 Geo + 0.25 RI	0.4906	711.1	**0.5825**	691.7	0.8111	0.9000	0.9921	1.000	0.4589	0.5020

Rarity-only performs best at 10 seconds but degrades at 60 seconds. Adding 25% rarity gives the best 60-second FVD, while adding RI gives the best 60-second LPIPS. Both blends reduce subject consistency, background consistency, aesthetics, and imaging quality relative to GeoCov. No composite dominates the complete metric suite, so the final method retains the simpler geometric score.

B Selected-Memory Quality

Table 8: Selected-memory quality on each policy's own 180-second rollout. Higher is better; "Late" is the final temporal portion.

Policy	Selected PSNR	Selected SSIM	Late PSNR	Late SSIM	Next PSNR	Next SSIM
Complete retention	11.703	0.3089	11.433	0.3146	11.308	0.3082
FIFO-32	10.619	0.2617	10.135	0.2546	10.113	0.2543
RI-32	13.396	0.3830	12.896	0.3723	11.323	0.3203
GeoCov-32	16.522	0.4759	15.119	0.4386	11.569	0.3253

Relative to complete retention, selected-frame differences are +1.693 dB PSNR and +0.0741 SSIM for RI, and +4.819 dB and +0.1670 for GeoCov. Late next-chunk SSIM improves by 0.0133 for RI (CI $[0.0028, 0.0279]$) and 0.0169 for GeoCov (CI $[0.0012, 0.0392]$); next-chunk PSNR intervals cross zero. This analysis is observational because earlier policy decisions change later pixels.

C Full Retention–Selection Sweep

Table 9: Common-source budget sweep on thirteen matched trajectories. Lower is better.

	B16		B32		B64		B128	
Policy	Ret.	Sel.	Ret.	Sel.	Ret.	Sel.	Ret.	Sel.
FIFO	.1872	.0448	.1668	.0646	.1429	.0867	.1203	.1084
RI	.0583	.1339	.0399	.1609	.0317	.1748	–	–
GeoCov	.0669	.1037	.0426	.1470	.0366	.1596	.0298	.1783

Complete retention has retention gap 0 and selection gap 0.2188. Every measured budget increase lowers retention gap and raises selection gap. RI-B128 is omitted because the common matched set lacks the required traces.

D Implementation Scaling

The decoded RGB archive occupies approximately 0.38 GB at 10 seconds and 2.30 GB at 60 seconds in the evaluated MemCam implementation. Extrapolations reach 22.7 GB at 10 minutes and 136 GB at 60 minutes. Corresponding cumulative exhaustive-retrieval estimates are 12 minutes, 20.9 hours, and 31.5 days. The long-duration values depend on the implementation and hardware and are not direct measurements. They illustrate bounded-state scaling, not a claim that exhaustive search is the only possible design.

E WorldMem Budget Sensitivity

Table 10: WorldMem 60-second LPIPS on the first fifteen matched videos. Complete retention obtains 0.652; lower is better.

Policy	B16	B32	B64	B128
FIFO	0.717	0.689	0.688	0.647
Latent-RI	0.566	0.546	0.549	0.567
Geometric Coverage	**0.525**	**0.534**	**0.545**	0.577
K-center	0.545	0.559	0.575	**0.559**
MCE	0.576	0.575	0.596	0.604

Selected matched FVD values at B32 are 1116.9 for Geometric Coverage, 1160.4 for Latent-RI, 3077.6 for complete retention, and 3554.9 for FIFO. Absolute WorldMem FVD is not compared with MemCam because the systems and evaluation sources differ.

F Reliability Probes

Table 11: Held-out online reliability signals. AUC predicts bottom-20% exact-index PSNR–SSIM fidelity.

Signal	AUC	Precision	Recall	Clean reject
Unclipped fraction	0.630	–	0.183	0.125
MUSIQ/PAQ2PIQ	0.516	–	0.314	0.307
Raw context similarity	0.546	0.294	0.245	0.151
Pose-conditioned residual	0.511	0.207	0.119	0.117
Anchor-chain score	0.463	–	0.000	0.000

Generic image-quality signals use only the generated frame; context and chain signals also use causal trace identity and camera geometry. Ground truth is used only for held-out labels and calibration. None of these signals is reliable enough to add to Eq. 8. A related offline observation also fails to produce a useful controller: older same-view memories are cleaner than later rewrites by 0.860 dB PSNR and 0.0397 SSIM over 823 pairs, but hard hysteresis worsens 60-second FVD from 690.5 to 768.7 and loses four VBench dimensions.

G Other Negative Results

Density-balanced coverage obtains LPIPS/FVD 0.5941/750.3 at 60 seconds versus 0.5850/690.5 for GeoCov. A 50/50 GeoCov–RI blend does not beat both constituents. MCE, trajectory coverage, and facility-style objectives also fail to improve the complete metric suite.

Several diagnostic hypotheses are unsupported. MemCam retrieves a fixed-size context before denoising, so the archive does not directly enlarge the denoiser's attention input. FOV-overlap hit rate is 100% at thresholds 0.01 and 0.1, with median selected overlap 0.9335, making frequent zero-overlap fallback an unlikely explanation. The largest repeated-frame share averages 0.172 and decreases with pool size ($\rho = -0.314$). Increasing Monte Carlo FOV sampling changes 39.1% of winners but yields mean appearance cost 0.0029, with fewer flips in larger pools. Finally, average archive corruption is not monotonic; deterioration is concentrated in the query-conditioned selected subset.

H Preliminary Context Replay

One valid replay holds the model, prompt, trajectory, seed, complete archive, and preceding contexts fixed, then substitutes GeoCov-selected context identities at one late section. LPIPS changes by -0.00177 and DINO distance by -0.01691. The direction is consistent with better conditioning, but $n = 1$ cannot support a causal claim. A multi-trajectory exact-index content replacement is required before using this result as primary evidence.

I Qualitative Memory Behavior

Figure 5 shows deliberately selected large disagreements, so aggregate conclusions should use Table 5. Figure 6 visualizes the local redundancy criterion; it does not claim that every retained substitute is intrinsically cleaner.

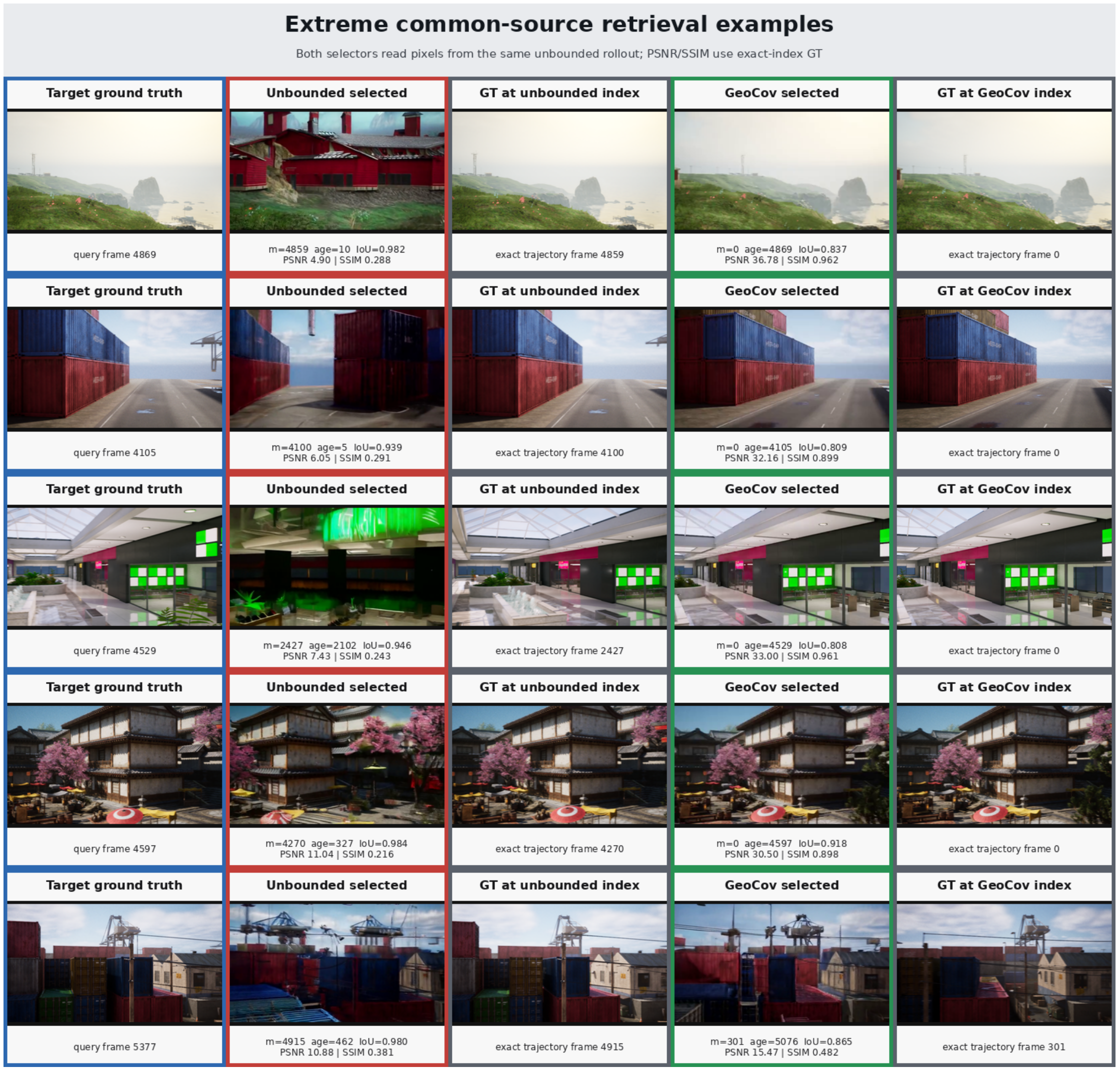

Figure 5: Large common-source retrieval disagreements. Candidate pixels come from one complete-retention rollout; policies supply only the selected indices. Cases were chosen using high-overlap and fidelity-gap filters and illustrate possibilities rather than frequency.

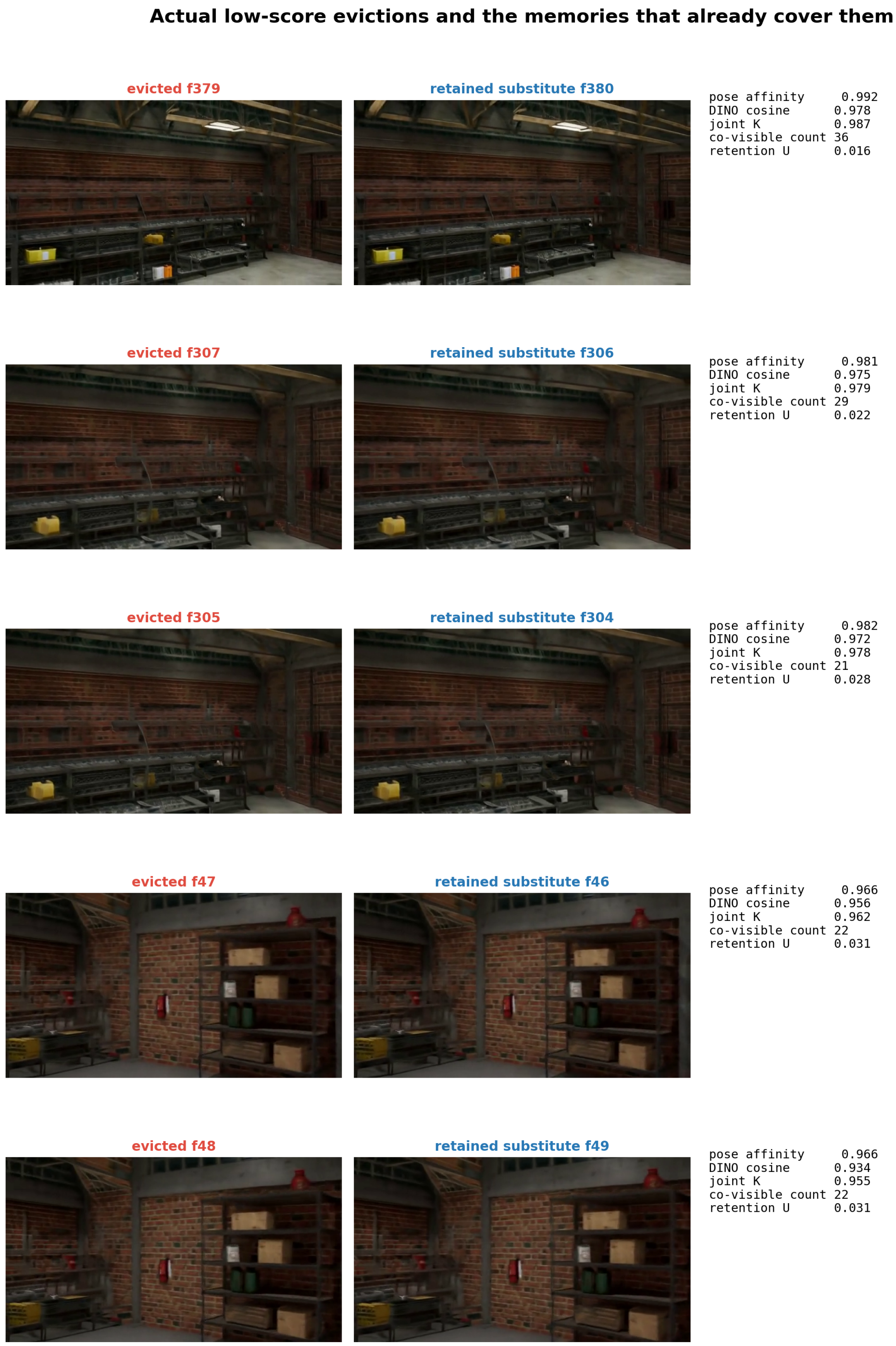

Figure 6: Recorded evictions and retained substitutes. In this Warehouse update, GeoCov reduces 108 candidates to 32. Each displayed eviction has high pose affinity and DINO similarity to a retained neighbor; recomputed and logged eviction sets overlap by 97.4%.

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